

CENTRAL UNIVERSITY OF KASHMIR

Tulmulla, Ganderbal, Jammu and Kashmir, 191131.



Masters in Technology (M Tech)
Computer Science & Engineering (CS&E)

COURSE STRUCTURE

NEP-2020

[BoS – 18/12/2024]

For Batch 2025

**School of Engineering & Technology
Department of Information Technology**

Course Structure for M.Tech. Computer Science & Engineering

Semester I					
S. No.	Course Code	Course Title	Credits	CIA	ESE
1	TAA.61.501	Object Oriented Software Engineering	4	50	50
2	TAA.61.502	Algorithms & Algorithmic Complexity	4	50	50
3	TAA.61.503	Algorithms & Algorithmic Complexity Lab	2	50	50
4	TAA.61.504	High-Performance Computer Architecture	4	50	50
5	TAA.61.505	High-Performance Computer Architecture Lab	2	50	50
6		Ability Enhancement Course-I	2	50	50
7		Value Added Course-I / Elective	2	50	50
		Total	20		
Semester II					
S. No.	Course Code	Course Title	Credits		
1	TAA.61.551	Next Generation Networks	4	50	50
2	TAA.61.552	Next Generation Networks Lab	2	50	50
3	TAA.61.553	Data Science	4	50	50
4	TAA.61.554	Data Science Lab	2	50	50
5		Elective /SEC-I /(MDC-I)	4	50	50
6		Ability Enhancement Course -II	2	50	50
7		Value Added Course –II / Elective	2	50	50
		Total	20		

Course Structure for M.Tech. Computer Science & Engineering

Semester III			
S. No.	Course Code	Course Title	Credits
1	TAA.61.601	Network Security	4
2	TAA.61.602	Network Security Lab	2
3	TAA.61.603	Research Methodology	2
4	TAA.61.604	Wireless communication	4
5	TAA.68.605	Research Project Phase I	4
6		Elective II/(MDC-II)	4
		Total	20

Semester IV			
S. No.	Course Code	Course Title	Credits
1	TAA.68.651	Research project Phase II (Dissertation)	18
2		Elective/Experiential Learning	02
		Total	20

Total Credits	80
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Course Structure for M.Tech. Computer Science & Engineering

Elective Basket for Semester II

S. No.	Course Code	Course Title	Credits
02	TAA.61.571	Internet of Things	4
03	TAA.61.572	Elective on Special Topics	4
04	TAA.61.573	Embedded Systems	4
05	TAA.61.574	Real-Time Operating System	4
06	TAA.61.575	Logic Programming	4
07	TAA.61.576	Advanced Graph Theory	4
08	TAA.61.577	Machine Learning	4
09		**MOOCs	

Elective Basket for Semester III

S. No.	Course Code	Course Title	Credits
01	TAA.61.621	Deep Learning	4
02	TAA.61.622	Operation Research	4
03	TAA.61.623	Foundations of LLM and Agentic AI	4
04	TAA.61.624	Cloud Computing	4
05	TAA.61.625	Robotics	4
06	TAA.61.626	Blockchain Technologies	4
07	TAA.61.627	Game Theory	4
08		**MOOCs	

Elective Basket for Semester IV

S. No.	Course Code	Course Title	Credits
01	TAA.61.671	Soft Computing	4
02	TAA.61.672	Evolutionary Computing	4
03	TAA.61.673	AI in Healthcare	4
04	TAA.61.674	DevOps for AI	4
05		**MOOCs	

****** Students may opt for MOOC (Massive Open Online Courses) as per the curriculum provisions, subject to the following conditions:

- i. Students may choose MOOC courses only with prior approval from the concerned department.
- ii. The selection of MOOC courses must comply with the maximum credit limit prescribed for MOOC courses under the programme regulations.
- iii. The department reserves the right to verify the academic relevance, equivalence, and suitability of the MOOC course before granting approval.
- iv. The decision of the department regarding approval or rejection of a MOOC course shall be final and binding.